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| Roll Number: _____                                                                                                        |            |                                                                   |
| <b>Thapar Institute of Engineering and Technology, Patiala</b><br>Department of Electronics and Communication Engineering |            |                                                                   |
| B. E. 2 <sup>nd</sup> year ENC & ENC(UoQ)                                                                                 | <b>MST</b> | Course Code: <b>UEC612</b>                                        |
| Course Name: <b>Digital System Design</b>                                                                                 |            | 7 <sup>th</sup> March, 2025                                       |
| Time: 2 Hours, M. Marks:30                                                                                                |            | Instructors: Dr. Alpana Agarwal, Dr. Manu Bansal & Dr. Anil Singh |

**Note: Attempt all part of a question at one place. Else, only first attempt will be evaluated.  
Make suitable assumptions, if required.**

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|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|----|
| 1. | A 12-bit Hamming code word containing 8 bits of data and 4 parity bits is read from memory. What was the original 8-bit data word that was written into memory if the 12-bit word read out is 000011101010.                                                                                                                                                                                                                                                                                   | 5 | CO1 | L2 |
| 2. | In a certain application, four inputs A, B, C, D (both True and Complement forms available) are fed to logic circuit, producing an output F, which operates a relay. The relay turns ON when $F(A,B,C,D) = 1$ for the input states (ABCD): '0000', '0010', '0101', '0110', '1101', and '1110'. States '1000' and '1001' do not occur, and for the remaining states, the relay is OFF. Minimize F with the help of a Karnaugh map and realize it using a minimum number of 3-input NAND gates. | 5 | CO1 | L2 |
| 3. | Two decimal numbers, both between 0 and 99, need to be added to each other. Design a logic circuit that performs a BCD addition of these two numbers. Use block diagrams for the binary adder(s). For the other components that you need, use any combinational logic gates that you require.                                                                                                                                                                                                 | 5 | CO2 | L3 |
| 4. | Design a 4:2 Priority Encoder with order of priority as $D_0 > D_1 > D_2 > D_3$ using logic gates.                                                                                                                                                                                                                                                                                                                                                                                            | 5 | CO2 | L2 |
| 5. | Implement the function $f(w1, w2, w3, w4) = \overline{w1}.\overline{w2}.\overline{w4} + w1.w2 + w1.w3 + w1.w4 + w3.w4$ by using a 4-to-1 multiplexer (use w1 on MSB and w4 on LSB select lines) and as few other gates as possible. Assume that only the True inputs w1, w2, w3, and w4 are available. Clearly draw the final circuit using standard symbols with proper labels.                                                                                                              | 5 | CO2 | L3 |
| 6. | (a) Tabulate the truth table for a ROM that implements the following Boolean functions:<br>$A(x, y, z) = \Sigma(0, 3, 4, 6)$<br>$B(x, y, z) = \Sigma(2, 3, 6, 7)$<br>$C(x, y, z) = \Sigma(1, 5)$<br>$D(x, y, z) = \Sigma(0, 1, 3, 5, 7)$<br>(b) What is the minimum size of ROM required for the implementation?<br>(c) Draw the symbolic schematic diagram for the implementation.                                                                                                           | 5 | CO2 | L2 |